



Unified AI Provider Platform (UAP) – Product Requirements Document

Background | Current State

Healthcare Revenue Cycle Management (RCM) is a complex domain involving many siloed processes—from patient registration and insurance eligibility checks to medical coding, claims submission, and denial handling. Today, various RCM teams are innovating with AI and automation (e.g., coding assistance tools, denial prediction models), but these efforts often operate in isolation.

Key Challenges in the Current State

- **Fragmented Data-Workflows:** Information is spread across systems. Without a unified platform, an AI model developed by one team (say, for coding) may not easily integrate with another team's process (like billing), leading to manual handoffs and incomplete context.
- **Slow Integration & Deployment:** Bringing a new model or automation into production is time-consuming. Each deployment may require custom IT work, causing months-long delays from development to real-world impact.
- **Duplicate Efforts:** In the absence of a central repository, different departments might unknowingly develop similar solutions (for example, two teams training models to predict denials).
- **Governance & Compliance Concerns:** Handling patient financial data and PHI with AI requires strict oversight. Today, there's no common governance framework for RCM AI initiatives—making leaders cautious.

Despite these challenges, our teams have made significant strides in applying AI to RCM. The opportunity now is to amplify and connect these successes by addressing the fragmentation and providing the support structure they need to flourish.

Opportunity

We have a chance to establish a unified platform that accelerates innovation for all RCM teams. Instead of each solution living in a silo, the **Unified AI Provider Platform (UAP)** will bring them together, enabling:

- **Rapid Time-to-Market:** UAP will drastically cut down the time it takes to get an AI agent or model from the lab into daily operations.
- **Bring Your Own Model (BYOM) Support:** UAP supports onboarding external models into the Model Marketplace with full governance and compliance checks.
- **Empowerment, Not Replacement:** UAP provides infrastructure so teams can focus on improving models rather than deployment logistics.
- **Adaptive & Proactive Operations:** By embedding AI agents at key points in the revenue cycle and coordinating them, UAP enables proactive management.
- **Unified Governance & Trust:** Centralized governance ensures compliance and builds trust in AI adoption.

Product Overview / Purpose

The **Unified AI Provider Platform (UAP)** is an enabling platform that will unify RCM operations with generative AI and intelligent automation. It serves as the central system for RCM AI initiatives, providing the underlying capabilities needed for disparate solutions to function as one cohesive, scalable system.

Primary Purpose

UAP's primary purpose is to help RCM teams deliver results faster and more reliably by offering:

- **Unified Data Access for AI:** UAP leverages the organization's existing data (e.g., the Common Data Platform) to ensure that every AI agent has the information it needs. By tapping into this single source of truth (without rebuilding it), all agents operate with full context (patient details, payer contracts, prior interactions, etc.), leading to more accurate and consistent outcomes.
- **"Model Marketplace" for AI Assets:** The platform provides a one-stop Model Marketplace where all AI models (internal or from partners) are registered, governed, and made available for reuse. This marketplace encourages sharing of innovations (so one team's model can benefit others), while enforcing standards—every model is documented, evaluated, and monitored centrally. It accelerates learning across the organization and avoids duplicate development.
- **MCP-Server Integration (Supervised/ML Models to Agentic Workflows)**

UAP exposes ML and supervised models from the Modeling module to agentic workflows through **MCP Servers**. These servers make model artifacts accessible as secure, governed endpoints that AI agents can call within workflows. This creates a seamless bridge between traditional supervised modeling efforts and the new agent-based automation layer—ensuring that proven predictive models can be operationalized instantly inside agentic processes without custom integration work.

- **Agentic Workflow Orchestration:** UAP introduces an agent orchestration layer that can coordinate multiple AI agents (and humans) in complex workflows. Think of each agent (coding assistant, eligibility checker, denial handler, etc.) as a member of a digital team—UAP is the manager that makes sure they communicate, hand off tasks, and collectively cover the end-to-end process. This orchestration is context-aware and can adapt in real time as conditions change.
- **One-Click Deployment & Scaling:** The platform simplifies deployment of AI solutions to a single action. Once a model or workflow is approved, it can be deployed with one click. This means RCM teams can update rules or roll out a new ML model quickly and safely.
- **Built-in Governance and Compliance:** UAP has governance at its core. It tracks versions, usage, and performance of each model and agent, and provides an audit trail for every automated action. Role-based access ensures that only authorized staff can change workflows or deploy models. This governance fabric gives confidence that even as we automate more, we remain compliant with regulations.
- **Ease of Use for Domain Experts:** Beyond the technical features, UAP is designed to be user-friendly for non-engineers. Through a visual Agent Builder interface, business users (RCM managers, analysts) can adjust workflows or create automation rules via drag-and-drop, without coding. This empowers the people who know the processes best to directly shape and refine them. It fosters a culture where operations and data science co-create solutions, rather than solutions being thrown over the wall.

In short, UAP is the glue and accelerator for our RCM AI efforts. It will connect the dots between data, people, and AI tools: using our enterprise data backbone, assembling a marketplace of intelligence, orchestrating agents into seamless workflows, and doing it all in a governed, rapid deployment manner. The ultimate purpose is to transform our revenue cycle into an intelligent, agile operation—with UAP enabling each team to contribute their piece of AI and have it immediately enhance the whole.

Personas

Group	Persona	Role Description
First (Internal)	Data Scientist	Builds, trains, evaluates internal models; conducts experiments.
	ML Engineer	Implements pipelines, deploys models, maintains training workflows.
	Platform Engineer	Manages platform infrastructure, performance, reliability.

	Data Engineer	Handles data ingestion, feature stores, and transformations.
	Product Manager	Defines platform roadmap, requirements, strategy.
	Governance/Compliance	Oversees responsible AI, approvals, audit workflows.
	Security Engineer	Ensures access controls, safety, isolation, secure integration.
	DevOps/SRE	Automates deployments, CI/CD, monitoring.
Second (Provider)	Provider Admin	Manages provider organization, user roles, platform configuration.
	Clinical Expert	Reviews model outputs for clinical relevance and safety.
	Domain Expert	Validates models from subject-matter perspective.
	Provider Data Scientist	Fine-tunes models with provider data; analyzes insights.
	Data/BI Analyst	Runs analytics, reporting, dashboards.
	IT/Security Staff	Handles integration, identity, compliance with provider systems.
	Operational Admin Staff	Uses workflow outputs and tools in daily operations.
	Review Board / Ethics Committee	Approves model usage, ensures clinical/ethical compliance.
Third (Ecosystem Partners)	Startup Science Team	Brings their own models; collaborates on experiments.
	Third-Party ML Engineer	Integrates models with platform APIs/frameworks.
	Vendor Engineer	Builds connectors, plugins, or extensions.
	ISVs	Independent companies offering software/models on the platform.
	External Auditor/Evaluator	Provides external validation, safety and compliance audits.

Scope | Use Cases

☒ Model Marketplace (with Governance) | Reusable AI Agent Library

The Model Marketplace is the central hub for all AI intelligence within UAP. It acts as a single source of truth for every model, prompt, and AI asset—whether developed internally or sourced externally from partners and vendors.

Requirements:

- **Central Repository:** Every model and prompt is registered with metadata (purpose, owner, version history).
- **Approval Workflow:** Models move through Draft → Review → Approved before production use.
- **Model Report Cards:** Easy-to-read evaluation summaries (accuracy, bias checks, limitations) for transparency.
- **Continuous Monitoring:** Detects performance drift or bias and triggers retraining or rollback. Provisioning Agent/Infrastructure to leverage external data sources for monitoring the model performance.
- **Champion vs. Challenger Testing:** Safely compare new models against current ones before full rollout.
- **BYOM (Bring Your Own Model):** External models from vendors or partners can be onboarded under the same governance standards.
- **MCP-Server:** All the models in the model marketplace should be exposed to the MCP-Server so that they can be used by the agentic workflow

Reusable AI Agent Library:

- **Pre-Built Agents:** Ready-to-use agents for common RCM tasks (Ex: eligibility checks, coding, denial prediction etc.)
- **Model Evaluation Agent:** Provisioning Agent to leverage external data sources for monitoring the model performance as needed (Also pointed in continuous monitoring)
- **Standardized Interfaces:** Consistent input/output formats for easy integration.
- **Continuous Updates:** Agents must automatically reference the most recent approved version of any model, ensuring they always operate on the latest improvements without requiring manual reconfiguration.
- **Extensible Design:** Teams can customize or add new logic without starting from scratch.

Why This Matters:

- **Accelerates Innovation:** Teams reuse proven models and agents instead of reinventing the wheel.
- **Governed Intelligence at Scale:** Every AI asset—internal or external—is vetted, monitored, and documented.
- **Safe Experimentation:** Champion–Challenger testing and rollback capabilities reduce risk.
- **Single Source of Truth:** Centralized repository prevents duplication and builds trust across the organization.
- **BYOM Advantage:** Opens the door to best-in-class external models while maintaining compliance and governance.

One-Click Deployment (CI/CD Pipeline)

The platform simplifies deployment of AI solutions to a single action. Once a model, agent, or workflow is approved, it can be deployed with one click. This means RCM teams can update rules or roll out a new ML model, agent, or prompt quickly and safely, accelerating innovation while maintaining compliance.

Requirements:

- **One-Click Deployment Across All AI Assets**
Enable instant deployment of approved ML models, supervised models, AI agents, prompts, MCP-served models, and workflows without manual intervention.
- **Unified Model Flexibility**
Agents and workflows must support seamless selection of different model types—pre-trained, fine-tuned, partner-provided, or custom (e.g., ChatGPT, Claude, internal LLMs)—during deployment.
- **Standardized I/O Contracts**
All deployable assets must adhere to standardized input/output schemas so they can plug into workflows, other agents, and external integrations consistently.
- **RAG Integration**
Assets may attach knowledge bases via RAG or access a standalone RAG module if used as a shared platform feature.
- **Search & Retrieval Capabilities**
Deployed agents and workflows should be able to perform secure online search, shared-folder search, or structured retrieval where applicable.
- **Inter-Agent Interoperability**
Agents must be able to call other agents, tools, or search history during execution, enabling composition of multi-agent workflows during deployment without extra configuration.

- **Zero-Downtime Deployment**

Use deployment strategies such as **blue-green**, **canary releases**, and **gradual rollout** to ensure updates reach production without interrupting ongoing RCM operations.

- **Monitoring, Alerts & Automated Rollback**

Track deployment status, runtime performance, latency, failure rates, and drift. If anomalies occur, the system should automatically roll back to a stable version.

- **Preview (Early Access) Mode**

Allow developers to deploy assets in **preview** to safely test new models, functions, or agent behaviors before GA. Preview mode should allow interactive testing (e.g., chatting with the agent) without affecting production flows.

Impact:

- **Faster Innovation:** Teams can push fixes and improvements in hours instead of weeks.
- **Continuous Improvement:** Frequent updates without fear of downtime or disruption.
- **Governance Built-In:** Every deployment follows compliance and audit rules, ensuring trust and accountability.

 Integrated Model Development Environment (IMDE)

Building and improving AI models can take too long—sometimes months—because teams lack easy access to compute resources and collaboration tools. IMDE solves this by creating a secure, production-like sandbox where data scientists and RCM experts can work together seamlessly.

Requirements:

- **Secure Sandbox with Unified Data:** IMDE provides a safe environment connected to enterprise data sources, so modelers can focus on experimentation.
- **Pre-Loaded Tools:** For not just AI-ML models but also for LLMs, multimodal and deep learning resources for developing complex models.
- **Collaboration & Version Control:** Teams can share notebooks, track changes, and maintain audit logs for transparency and collaborate with domain experts for further insights and feedback.
- **One-Command Push to Model Marketplace:** Once a model is ready, it can be published to the marketplace for evaluation and deployment with a single command.
- **Continuous Improvement Loop:** IMDE makes it easy to pull recent real-world cases where models struggle, retrain, and push updates back to production quickly.

Impact:

- **Accelerates Innovation:** Self-service access to data and compute reduces model development from months to days.
- **Improves Model Quality:** Models are developed in a production-like setting, reducing the risk of “works in lab but fails in production.”
- **Empowers Domain Experts:** Billing and coding experts can collaborate with data scientists to test hypotheses and improve accuracy.
- **Continuous Learning:** Every payer change or trend becomes an opportunity for immediate learning, not a delayed problem.

Dynamic Agent Orchestrator

UAP introduces an agent orchestration layer that can coordinate multiple AI agents (and humans) in complex workflows. Think of each agent (coding assistant, eligibility checker, denial handler, etc.) as a member of a digital team—UAP is the manager that makes sure they communicate, hand off tasks, and collectively cover the end-to-end process. This orchestration is context-aware and can adapt in real time as conditions change.

Requirements:

- **Typed Node Framework:** Ability to compose workflows using well-defined node types (Agent, MCP, Model, Tool, Control-Flow, Human-in-the-Loop, Governance).
- **Context Sharing:** All agents read/write from a unified claim context so decisions stay consistent across steps.
- **Event-Driven Coordination:** Agents trigger each other based on business events (e.g., denial received, auth completed).
- **Adaptive Decision Logic:** Support conditional routing, exception paths, and compensation logic.
- **Governance & Safety:** Enforcement of PHI/PII policies, RBAC/ABAC controls, audit logs, and rollback.
- **Human Escalation:** Ability to route to human review when confidence is low or compliance requires oversight.
- **Observability Hooks:** Built-in metrics, traceability, and outcome evaluation across all steps.

Impact:

- **End-to-End Automation:** Eliminates manual handoffs and creates a smooth flow from claim creation to resolution.
- **Nothing Gets Missed:** Every denial is followed up, every claim gets resolved—no duplicate work.
- **Prioritization:** High-value or urgent claims are handled first, reducing revenue risk.
- **Audit Trail:** Complete visibility into agent and human actions for compliance.

- **Graceful Exception Handling:** Automation pauses when needed and involves humans, avoiding silent failures.

Visual Agent Builder UI

A **Visual Agent Builder UI** gives non-engineers (RCM managers, analysts) the power to design, adjust, and test workflows through an intuitive drag-and-drop interface. This democratizes automation, reduces IT dependency, and ensures the people closest to the process can continuously refine and optimize it.

1. Drag-and-Drop Workflow Assembly

- Users can visually place steps like *Eligibility Check*, *Run ML Model*, *Denial Review*, *Human Approval*, etc.
- Supports branching logic (If/Else), routing rules, and workflow sequencing without writing code.

2. Built-In Guardrails & Compliance Protections

- Prevents unsafe actions (PHI in emails, infinite loops, unauthorized overrides).
- Enforces RBAC/ABAC so sensitive configurations are only accessible to approved users.

3. Telemetry, Debugging & Test Runs

- Ability to “test-run” a single claim to see how the workflow behaves.
- Users can attach metrics, logs, and alerts to specific steps for easier troubleshooting.

4. Versioning, Approvals & Rollback

- Every workflow saved with full history.
- Requires approval before changes go live.
- One-click rollback to a previous version if issues arise.

5. Reusable Subflows & Templates

- Prebuilt workflows for common patterns (Appeal Handling, Manager Notification, Auth Checks).
- Enables reuse to keep processes consistent across teams.

6. Interoperability & Extensibility

- Standard schemas to connect any agent, model, or external service.
- Advanced users can add custom logic with safeguards.

7. Agent Attribute Configuration at Build/Publish Time

Each agent node exposes configuration options so the business can explicitly choose:

- **Model Choice:** e.g., ChatGPT, Claude, internal LLM versions.
- **Input & Output Schema:** define what the agent receives and returns.
- **Knowledge Source (RAG):** attach or swap knowledge bases.
- **Online Search Capability:** toggle search on/off with domain restrictions.
- **Additional Agent Settings:** temperature, max tokens, tool permissions, and safety settings (hallucination guardrails, PHI masking, etc.).

Impact:

- **Rapid Iteration:** Analysts can tweak workflows in minutes instead of waiting for IT or vendor patches, reducing denial spikes and keeping cash flow steady.
- **Empowers Domain Experts:** People who know the revenue cycle best can contribute directly without coding, while compliance guardrails keep everything safe.
- **Consistency & Best Practices:** One central place for workflows means uniform standards across departments and easy audits.
- **Transparency & Visibility:** Visual workflows make automation understandable. Supervisors can see where bottlenecks occur and fix them quickly.
- **Lower Maintenance & Easier Onboarding:** Changes are safer and easier, reducing reliance on technical teams and lowering long-term costs.

Non-Functional Requirements (Summary)

Category	Key Points
Performance	Scalable for high claim volumes; low latency, real-time processing
Reliability	99.9% uptime
Security	Role-based access; PHI/PII masking; full audit logs; HIPAA/SOC2 compliance
Maintainability	Modular architecture; version control & rollback; auto documentation
Usability	Intuitive low-code UI; real-time dashboards; accessibility (WCAG)

Interoperability	Seamless integration with RCM/CDP; standardized APIs
Observability	Unified monitoring & alerts; KPI tracking
Extensibility	Safe customizations; plugin support for third-party models

Success Metrics for UAP

Metric	Description	Example Target
Time to Market	Time from model/workflow readiness to production deployment	X% of approved changes deployed within 1 week
Model & Agent Reuse Rate	Measures reuse of models/agents across teams	Ratio of deployed vs. developed models > 1
Platform Adoption	Portion of RCM workflows handled by UAP	UAP involved in Y% of claims by year-end
User Engagement & Self-Service	Tracks business user empowerment (workflow edits, rule additions)	X workflow changes by analysts without IT intervention
Audit & Compliance	Ensures governance and zero rogue scripts	100% models have approved model cards; near-zero incidents
Team Productivity/Capacity	Reduction in grunt work for IT/Data Science teams	Data scientists spend x% on modeling vs. y% before

Appendix

Persona Summary – Table Format

High-Level → Persona Mapping

Group	Persona Category	Specific Personas
First (Internal)	Technical	Data Scientists, ML Engineers, Platform Engineers, Data Engineers
	Product & Governance	Product Managers, Compliance/Governance Teams, Security Engineers
	Operations	DevOps/SRE, Infra Admins
Second (Provider)	Administrative	Provider Admin, Operational Admin Staff
	Domain/Clinical	Clinical Experts, Domain Experts
	Technical	Provider Data Scientists, Analysts, IT Staff
	Oversight	Review Board, Ethics Committee
Third (Ecosystem Partners)	Technical	Startup DS Teams, ML Engineers, Vendor Engineers
	Commercial/Partner	ISVs (Independent Software Vendors)
	External Governance	External Auditors, External Evaluators